Security Operations Centre Incident Investigation Report

Scenario 2 - Sysmon Telemetry and Wazuh Alerts

Evidence sources	Scenario_2_Alerts.json; Scenario_2_Archives.json
Host	DESKTOP-5OG7C28 / Wazuh agent Windows_VM (192.168.232.130)
Primary user	DESKTOP-5OG7C28\Darragh Casey
Investigation window	2026-10-03 16:21:19.986 UTC to 16:31:03.350 UTC
Primary activity window	2026-10-03 16:21:19.986 UTC to 16:26:15.459 UTC
Telemetry reviewed	160 archive records and 15 Wazuh alert records
Time convention	Timeline uses event UTC timestamps; Wazuh alert timestamps are recorded at UTC+02:00

Assessment basis: conclusions are derived from the supplied telemetry and alerts. Alert labels were not treated as proof of malicious activity. Observed facts and inference are separated throughout.

1. Executive Summary

The evidence shows a concentrated local discovery sequence executed on DESKTOP-5OG7C28 under the user Darragh Casey from an already-running high-integrity PowerShell session. At 16:26:15 UTC, PowerShell PID 5664 created cmd.exe PID 11280 with a chained command line that invoked local-account enumeration, directory listing of C:\Users, Credential Manager enumeration, and local-group enumeration. Child process telemetry confirms execution of net.exe/net1.exe and cmdkey.exe, while Windows Security events independently confirm Credential Manager enumeration and enumeration of the Builtin Users group.

Approximately 2 minutes 27 seconds earlier, Notepad.exe created C:\Users\Darragh Casey\Desktop\T1087.001\Test_2.ps1. The path and timing make the script a plausible precursor to the later discovery activity, especially in the stated isolated laboratory context. However, the supplied telemetry does not contain the script body and does not show a direct PowerShell process creation with a -File or equivalent argument referencing Test_2.ps1. Accordingly, execution of that specific script is inferred rather than directly observed.

Several Wazuh alerts were not substantiated as malicious when their underlying records were assessed. The dllhost.exe alert contains a standard System32 COM Surrogate invocation and no corroborating injection telemetry. The BITS startup-type changes occur amid Windows maintenance/update activity and are later reversed. The alert for an executable written beneath C:\Windows\SoftwareDistribution is directly followed by Microsoft Defender update execution and successful update events, supporting a benign update explanation rather than lateral tool transfer.

Overall, the supported security-relevant activity is best characterized as elevated local host reconnaissance and credential-store enumeration in a lab exercise. There is no supplied evidence of persistence, command-and-control, lateral movement, destructive impact, account creation, privilege-escalation mechanism, or extraction of credential secrets. Overall investigation confidence: High.

2. Scope of Activity

The investigation covered all 160 records in Scenario_2_Archives.json and all 15 records in Scenario_2_Alerts.json. The archive includes Sysmon process, file, network, DNS, and registry telemetry together with Windows Security and service/update events. The alert file contains Wazuh detections derived from a subset of those events.

The evidence concerns one Windows host, DESKTOP-5OG7C28, monitored as Wazuh agent Windows_VM at 192.168.232.130. The principal interactive user in the relevant records is DESKTOP-5OG7C28\Darragh Casey. Two user logon contexts are visible: a high-integrity context using Logon ID 0xA8D6D and a medium-integrity context using Logon ID 0xA8DA9.

The primary investigative scope is the high-integrity PowerShell-led activity from 16:21:19 UTC through 16:26:15 UTC, including repeated hostname/whoami executions and the later cmd.exe discovery chain. The .ps1 file creation at 16:23:47 UTC is retained as potentially related context. Windows maintenance, Defender update, OneDrive update, Edge update, Software Protection, and routine service events were reviewed for possible relationships but excluded where the telemetry did not support linkage to the discovery chain.

Important limitations: the excerpt begins after PowerShell PID 5664 was already running, so its original launch mechanism is not present; the content of Test_2.ps1 is unavailable; command output is not captured; and the telemetry does not establish what information, if any, was subsequently used. No evidence outside the supplied files was used to determine the scenario outcome.

3. Detailed Investigation

3.1 High-integrity PowerShell discovery context

Observed: Multiple Sysmon Event ID 1 records show HOSTNAME.EXE and whoami.exe launched as direct children of C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe, PID 5664, under Darragh Casey with IntegrityLevel High and Logon ID 0xA8D6D. This pattern appears at 16:21:19-16:22:30 UTC and again immediately before the main 16:26:15 UTC discovery chain. Examples are archive lines 1-6, 12-13, and 102-107.

Inference: hostname and whoami are consistent with host and user discovery. Because they repeat under the same long-running PowerShell process, they may represent deliberate reconnaissance or commands executed as part of an interactive/test workflow. The telemetry does not contain their output and does not prove why they were repeatedly launched, so they are treated as supporting rather than decisive evidence.

3.2 Creation of a PowerShell script artifact

Observed: Sysmon Event ID 11 records at 16:23:47.871 and 16:23:48.051 UTC show Notepad.exe writing the same target, C:\Users\Darragh Casey\Desktop\T1087.001\Test_2.ps1. Both records use the same ProcessGuid and the same CreationUtcTime, so they are treated as repeated telemetry for one file artifact rather than two independent scripts. These are archive lines 92-93 and Wazuh alert lines 3-4.

Observed: The exact Notepad process that performed the write is recorded as PID 10420, but a corresponding process-creation record for PID 10420 is not present in the supplied archive excerpt. A different Notepad process creation (PID 7756) appears earlier at archive line 21, so the report does not assume those PIDs are the same process.

Inference: The directory name T1087.001 and the later local-account discovery behavior make Test_2.ps1 a plausible lab test artifact related to the activity. The evidence does not contain the script contents or a direct process-creation event showing that file being passed to PowerShell, so the report does not state that Test_2.ps1 was executed as fact.

3.3 Main command-shell discovery chain

Observed: At 16:26:15.192 UTC, high-integrity PowerShell PID 5664 created cmd.exe PID 11280 with the command line shown below. The process inherited Darragh Casey, Logon ID 0xA8D6D, TerminalSessionId 1, and High integrity. The working directory was the user's Temp directory.

```
"cmd.exe" /c net user & dir c:\Users\ & cmdkey.exe /list & net localgroup "Users" & net localgroup
```

Observed: The ampersand-separated command line expresses a sequential discovery workflow. Sysmon then records child processes for net.exe user, net1.exe user, cmdkey.exe /list, net.exe localgroup "Users", net1.exe localgroup "Users", net.exe localgroup, and net1.exe localgroup. All execute in the same high-integrity user context and within approximately 267 milliseconds.

Observed: Windows Security Event ID 5379 at 16:26:15.380 UTC records 'Credential Manager credentials were read' with Read Operation 'Enumerate Credentials' by Darragh Casey under Logon ID 0xA8D6D. This independently corroborates cmdkey.exe /list. Windows Security Event ID 4799 at 16:26:15.434 UTC records enumeration of Builtin\Users by C:\Windows\System32\net1.exe, independently corroborating the local-group command.

Inference: The chain is reconnaissance-oriented. net user targets local account discovery; dir C:\Users reveals local profile directories; cmdkey /list enumerates stored Credential Manager entries; net localgroup "Users" enumerates membership of a specific local group; and net localgroup enumerates available local groups. The commands gather host/user authorization context rather than modifying it.

3.4 Privilege context

Observed: The PowerShell child processes, cmd.exe, net.exe/net1.exe, and cmdkey.exe all run with IntegrityLevel High in Logon ID 0xA8D6D. The medium-integrity Notepad/script-write activity is associated with a different logon context, 0xA8DA9.

Inference: The discovery sequence therefore had elevated access relative to a normal medium-integrity desktop process. However, the evidence does not show how the PowerShell session became elevated. No privilege-escalation technique should be inferred solely from the High integrity label.

3.5 Assessment of Wazuh alerts not retained as incident activity

dllhost.exe alert (Wazuh rule 61638, alert line 1): Observed telemetry shows C:\Windows\System32\dllhost.exe, Microsoft metadata, Medium integrity, and the conventional /Processid:{GUID} COM Surrogate argument. The parent image is unavailable, but the archive contains no Sysmon remote-thread, process-access, or other corroborating injection event tied to this ProcessGuid. Conclusion: the rule label 'Process Injection' is not supported by the supplied underlying evidence. The event is excluded from the main chain.

BITS service-change alerts (rule 61104, alert lines 2 and 6): The Background Intelligent Transfer Service start type changes from demand to auto at 16:23:00 UTC and back to demand at 16:25:05 UTC. The surrounding archive contains UsoClient, MoUsoCoreWorker, TrustedInstaller/TiWorker, PushToInstall, and other maintenance activity. No user discovery process is shown changing BITS and the setting is restored. Conclusion: insufficient evidence of adversarial persistence or transfer; excluded as unrelated system maintenance.

Software Protection scheduling alert (rule 60642, alert line 5): The underlying Application event states the Software Protection service was successfully scheduled for restart with reason RulesEngine. No relationship to the PowerShell

discovery chain is present. Excluded as routine service activity.

Executable-drop alert (rule 92217, alert line 15): Sysmon records svchost.exe writing C:\Windows\SoftwareDistribution\Download\Install\AM_Delta_Patch_1.459.518.0.exe as SYSTEM. Within the same update sequence, wuaucltcore.exe launches the patch, MpSigStub.exe handles the payload, Windows Update reports installation of a Microsoft Defender security-intelligence update, and a later event reports successful installation. Conclusion: this is strongly supported as legitimate Defender/Windows Update activity, not lateral tool transfer.

Script-creation alert (rule 92200, alert lines 3-4): The underlying file creation is real and relevant as a potential lab artifact, but the alert's Ingress Tool Transfer mapping is not independently supported because the file was written locally by Notepad.exe and no network transfer source is shown.

net.exe alert mapping (rule 92036, alert lines 8, 11, and 14): The Windows Command Shell aspect is supported, but the alert also labels DLL Search Order Hijacking. No DLL-load or search-order manipulation evidence is present, so T1574.001 is not supported by the supplied telemetry.

4. Evidence Analysis

The strongest evidence is the parent-child process lineage and matching security-audit events. The table below lists the records that directly establish the main behavior. Archive line numbers refer to Scenario_2_Archives.json; alert line numbers refer to Scenario_2_Alerts.json.

Evidence	Key values	Assessment
Archive L108 / Alert L7	Sysmon EID 1; cmd.exe PID 11280; parent PowerShell PID 5664; High; Logon 0xA8D6D; full chained command line	Primary execution record establishing the discovery workflow.
Archive L109-L110 / Alerts L8-L9	net.exe -> net1.exe; command 'net user'; same high-integrity logon	Direct local account discovery process chain.
Archive L111-L112 / Alert L10	cmdkey.exe /list; Security EID 5379 'Enumerate Credentials'	Direct process plus independent audit confirmation of Credential Manager enumeration.
Archive L114-L117 / Alerts L11-L14	net.exe/net1.exe localgroup 'Users' and localgroup; same parent cmd.exe	Direct local permission-group discovery.
Archive L113	Security EID 4799; Builtin Users; Process Name net1.exe	Independent confirmation of local-group membership enumeration.
Archive L92-L93 / Alerts L3-L4	Sysmon EID 11; Notepad.exe -> Desktop\T1087.001\Test_2.ps1	Observed script artifact creation; execution of this file not proven.
Archive L1-L6, L12-L13, L102-L107	HOSTNAME.EXE and whoami.exe; parent PowerShell PID 5664; High	Supporting host/user discovery context under the same PowerShell session.
Archive L127-L133 / Alert L15	SoftwareDistribution Defender patch; wuaucltcore.exe; MpSigStub.exe; Windows Update success	Corroborates benign update activity and excludes alert from incident chain.

4.1 Process lineage

The process genealogy is internally consistent: PowerShell PID 5664 -> cmd.exe PID 11280 -> net.exe/cmdkey.exe -> net1.exe where applicable. ProcessGuid and ParentProcessGuid fields match across the child records. This is stronger evidence than the alert labels because it shows exactly which commands were executed and by which parent.

4.2 User and logon correlation

The main chain consistently records User DESKTOP-5OG7C28\Darragh Casey, Logon ID 0xA8D6D, TerminalSessionId 1, and High integrity. Security audit events 5379 and 4799 use the same subject and Logon ID, linking the audit records to the process activity. The .ps1 write uses the same named user but a separate medium-integrity logon context (0xA8DA9), which prevents claiming process continuity solely from user identity.

4.3 Network and follow-on activity

The archive contains only one Sysmon network-connection record in the supplied window, and it belongs to OneDriveStandaloneUpdater.exe connecting to 52.123.128.14:443 after resolving ecs.office.com. No network connection

is attributed to PowerShell PID 5664, cmd.exe PID 11280, net.exe/net1.exe, or cmdkey.exe. No supplied evidence therefore supports command-and-control or data exfiltration by the discovery chain.

The archive also contains no supported account-creation, group-modification, service-installation, scheduled-task creation by the user discovery chain, or destructive action. This constrains the observed activity to discovery and credential-store enumeration rather than persistence or impact.

5. Timeline of Events

Times below are normalized to UTC using the event timestamps in the underlying Windows/Sysmon records.

UTC time	Event	Evidence / interpretation
16:21:19.986	HOSTNAME.EXE from PowerShell PID 5664	High-integrity user context begins within supplied excerpt. Archive L1.
16:21:20.032-16:22:30.332	Repeated HOSTNAME.EXE and whoami.exe	Same PowerShell PID 5664 and Logon 0xA8D6D. Supporting discovery context. Archive L2-L6, L12-L13.
16:22:44.804	dllhost.exe COM Surrogate	Wazuh rule 61638 fires, but no corroborating injection evidence. Archive L14 / Alert L1.
16:23:00.186	BITS start type demand -> auto	Occurs amid system update/maintenance activity. Archive L63 / Alert L2.
16:23:47.871	Test_2.ps1 created by Notepad.exe	C:\Users\Darragh Casey\Desktop\T1087.001\Test_2.ps1. Archive L92-L93 / Alerts L3-L4.
16:25:05.047	BITS start type auto -> demand	Previous service change is reversed. Archive L101 / Alert L6.
16:26:14.369-16:26:14.806	More HOSTNAME.EXE / whoami.exe	Same high-integrity PowerShell parent immediately before main chain. Archive L102-L107.
16:26:15.192	PowerShell -> cmd.exe discovery chain	cmd /c net user & dir C:\Users\ & cmdkey /list & net localgroup "Users" & net localgroup. Archive L108 / Alert L7.
16:26:15.264-16:26:15.295	net.exe / net1.exe user	Local account enumeration. Archive L109-L110 / Alerts L8-L9.
16:26:15.363-16:26:15.380	cmdkey.exe /list and EID 5379	Credential Manager entries enumerated; no proof that secret values were extracted. Archive L111-L112 / Alert L10.
16:26:15.387-16:26:15.459	net localgroup commands	Users membership and local groups enumerated; EID 4799 confirms Builtin Users enumeration. Archive L113-L117 / Alerts L11-L14.
16:27:15.475-16:27:27.964	Microsoft Defender update sequence	SoftwareDistribution patch, MpSigStub, update start and success. Supports benign explanation for Alert L15. Archive L127-L133.

6. MITRE ATT&CK Mapping

Mappings below are based on the observed command behavior, not on Wazuh's rule metadata. Confidence reflects how directly the supplied telemetry supports each technique.

Tactic	Technique	Evidence	Confidence
Execution	T1059.003 - Windows Command Shell	cmd.exe /c executes the chained discovery commands; child process lineage confirms execution.	High
Discovery	T1087.001 - Account Discovery: Local Account	net user / net1 user execute without a domain argument in the local host context.	High
Discovery	T1069.001 - Permission Groups Discovery: Local Groups	net localgroup "Users" and net localgroup; Security EID 4799 confirms Builtin Users membership enumeration.	High
Discovery	T1083 - File and Directory Discovery	dir C:\Users\ is explicitly present in the cmd.exe command line, indicating enumeration of local user profile directories.	High
Credential Access	T1555.004 - Credentials from Password Stores: Windows Credential Manager	cmdkey.exe /list plus Security EID 5379 'Enumerate Credentials'. The evidence supports enumeration of stored credential entries, not extraction of secret password material.	Medium
Discovery	T1033 - System Owner/User Discovery	Repeated whoami.exe execution as a child of the same elevated PowerShell session.	Medium

Tactic	Technique	Evidence	Confidence
Discovery	T1082 - System Information Discovery	Repeated HOSTNAME.EXE execution under the same elevated PowerShell session provides host identity information.	Medium
Execution	T1059.001 - PowerShell	PowerShell PID 5664 is the direct parent of the main cmd.exe chain and repeated discovery utilities. The excerpt does not contain its creation event or prove execution of Test_2.ps1.	Medium

Mappings explicitly not supported by the underlying evidence: T1055 Process Injection for the dllhost.exe alert; T1105 Ingress Tool Transfer for the locally created .ps1 file; T1574.001 DLL Search Order Hijacking for net.exe child processes; and T1570 Lateral Tool Transfer for the Defender update executable. The corresponding alert rules may be useful detection heuristics, but the supplied records do not establish those techniques in this incident.

7. Overall Investigation Conclusion

Observed facts: A high-integrity PowerShell process belonging to Darragh Casey spawned repeated hostname/whoami utilities and later created a high-integrity cmd.exe process containing a chained local discovery command. The chain executed net user, cmdkey /list, and localgroup commands through net.exe/net1.exe; Security auditing independently recorded Credential Manager enumeration and Builtin Users group enumeration. A .ps1 file named Test_2.ps1 was created in a Desktop\T1087.001 directory by Notepad before the main chain. No suspicious network communication by the discovery processes, no account or group modification, no persistence action tied to the user chain, and no destructive action are present in the supplied records.

Inferred conclusions: The most likely interpretation, taking the stated isolated-lab context into account, is deliberate execution of a local reconnaissance test from an elevated PowerShell environment. The Test_2.ps1 artifact is likely related to that test, but the evidence is insufficient to prove that this exact file was executed. The activity would be security-significant on a production endpoint because it enumerates accounts, profile directories, local groups, and stored Credential Manager entries from an elevated context.

The Wazuh alerts should not be read literally as a single multi-stage attack. Several high- or medium-severity detections are better explained by ordinary Windows component behavior or update activity after examination of their source events. The defensible incident finding is therefore narrower: elevated local discovery and credential-store enumeration, with no supplied evidence of subsequent exploitation, persistence, lateral movement, command-and-control, or impact.

Overall confidence: High.

Confidence is high because the principal command line, process lineage, user/logon context, and corroborating Security events are directly observed. Confidence is reduced only for the relationship between Test_2.ps1 and the later execution chain, the purpose of the repeated hostname/whoami calls, and any actions outside the captured telemetry window.